

Lab 1

Launch Template Instance Details:

A2-Lab1 (lt-097fb09a35de5f9c2)

Launch template details

Launch template ID

lt-097fb09a35de5f9c2

Launch template name

A2-Lab1

Default version

3

Owner

arn:aws:iam::865268032810:root

Details

Versions

Template tags

Launch template version details

Version

3 (Default)

Description

-

Date created

2026-02-18T19:55:14.000Z

Created by

arn:aws:iam::865268032810:root

Action

Instance details

Storage

Resource tags

Network interfaces

Advanced details

AMI ID

ami-05efc83cb5512477c

Instance type

t3.micro

Availability Zone

-

Availability Zone Id

-

Key pair name

a2-main

Security groups

sg-0e418bcea2e77bef4

Launch Template Advanced Details:

Launch template version details

Version

3 (Default)

Description

-

Date created

2026-02-18T19:55:14.000Z

Created by

arn:aws:iam::865268032810:root

Actions

Delete template version

Instance details

Storage

Resource tags

Network interfaces

Advanced details

Spot purchasing options

IAM instance profile

-

Stop-hibernate behavior

-

Resource-based IPv4 DNS

-

Placement group

-

EBS optimized instance

-

Tenancy affinity

-

License configurations

-

AMD SEV-SNP

-

Metadata IPv6 endpoint

-

User data

#!/bin/bash

set -e

----- System setup -----

yum update -y

yum install -y python3 openssl

APP_DIR=/home/ec2-user/https-server

mkdir -p \$APP_DIR

cd \$APP_DIR

----- Generate self-signed TLS certificate -----

openssl req -x509 -newkey rsa:2048 \

-keyout server.key \

-out server.crt \

-days 365 \

Instance auto-recovery

-

Termination protection

-

Resource-based IPv6 DNS

-

Target partition

-

Tenancy

-

RAM disk ID

-

Core count

-

Metadata accessible

-

Allow tags in metadata

-

Shutdown behavior

-

Stop protection

-

Detailed CloudWatch monitoring

-

Capacity reservation preference

-

Tenancy host resource group

-

Kernel ID

-

Threads per core

-

Token hop limit

-

Instance Configurable Bandwidth

-

Purchasing option

-

Hostname type

-

T2/T3 Unlimited

-

Capacity reservation target

-

Tenancy host ID

-

Enclave

-

Nested virtualization

-

Metadata version

-

Security Group:

sg-0e418bcea2e77bef4 - assignment2-l1

Actions

Details

Security group name

assignment2-l1

Security group ID

sg-0e418bcea2e77bef4

Description

Lab 1 for CS459

VPC ID

vpc-0a7bf477be5a3a626

Owner

865268032810

Inbound rules count

2 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (2)

Manage tags

Edit inbound rules

Search

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-05a44708751e4d52b	IPv4	SSH	TCP	22	206.127.179.198/32	-
☐	-	sgr-0afef20207bc6e89d	IPv4	Custom TCP	TCP	8443	0.0.0.0/0	-

a2-l1

a2-l1 Capacity overview

am:aws:autoScaling-us-east-2:865268032810:autoScalingGroup:c9f8923f-7b9d-47b4-a28d-29876aa54853:autoScalingGroupName/a2-l1

Desired capacity 2	Scaling limits 2 - 3	Desired capacity type Units (number of instances)	Status -
-----------------------	-------------------------	--	-------------

Date created
Wed Feb 18 2026 14:13:04 GMT-0600 (Central Standard Time)

[Edit](#)

Details | Integrations | Automatic scaling | Instance management | Instance refresh | Activity | Monitoring | Tags

Launch template

Launch template lt-O97fb09a35de5f9c2 A2-Lab1	AMI ID ami-05efc83cb5512477c	Instance type t3.micro	Owner arn:aws:iam::865268032810:root
Version Default	Security groups -	Security group IDs sg-0e418bcea2e77bef4	Create time Wed Feb 18 2026 13:55:14 GMT-0600 (Central Standard Time)
Description -	Storage (volumes) -	Key pair name a2-main	Request Spot Instances No

[View details in the launch template console](#)

[Edit](#)

Network

Availability Zones us2-az2 (us-east-2b) us2-az3 (us-east-2c) us2-az1 (us-east-2a)	Subnet ID subnet-0c8db012a73e38d4a subnet-09e34afdadcf36320 subnet-0d2f7b78048b855e61	Availability Zone distribution Balanced best effort
---	---	---

Instance type requirements

Your Auto Scaling group adheres to the launch template for purchase option and instance type.

Health checks

Health check type EC2	Health check grace period 300
---------------------------------	---

A2-L1

Actions

Details

arn:aws:elasticloadbalancing:us-east-2:865268032810:targetgroup/A2-L1/d8f77db96b361417

Target type
Instance

Protocol : Port
TCP: 8443

VPC
[vpc-0a7bf477be5a3a626](#)

IP address type
IPv4

Load balancer
[A2-Lab1](#)

Total targets	Healthy	Unhealthy	Unused	Initial	Draining
2	2	0	0	0	0

► **Distribution of targets by Availability Zone (AZ)**

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Health check settings

Protocol
TCP

Port
Traffic port

Timeout
10 seconds

Unhealthy threshold
2 consecutive health check failures

Healthy threshold
5 consecutive health check successes

Interval
30 seconds

Edit

Load Balancer Listener Tab:

Listeners (1)

A listener checks for connection requests using the protocol and port that you configure. Traffic received by a Network Load Balancer listener is forwarded to the selected target group.

Filter listeners							
☐	Protocol:Port	Default action	ARN	Security policy	Default SSL/TLS certificate	ALPN policy	Tags
☐	TCP:8443	Forward to target group A2-L1 1 (100%) Target group stickiness: Off	ARN	Not applicable	Not applicable	None	0 tags

Output of curl -k https://<load-balancer-DNS>:8443

```
PS C:\Users\evan\OneDrive\CODE\CS459\assignment-2> curl.exe -k https://A2-Lab1-37c93771f87b6c1f.elb.us-east-2.amazonaws.com:8443
HTTPS Echo Service

Hostname: ip-172-31-19-152.us-east-2.compute.internal
IP Address: 172.31.19.152
Timestamp (UTC): 2026-02-19T02:13:48.075794
```

Lab 2